

UBC

Electrical and Computer Engineering

**Modeling and Interface-Circuit Optimization
for Piezoelectric Energy Harvesting: From
Vibration Benchmarks to Low-Voltage
Underwater Sources**

Undergraduate Thesis

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Abstract

This work investigates one continuous question across progressively more realistic circuit models: how the electrical interface of a piezoelectric energy harvester can be modeled, validated, and optimized when the available source changes from a conventional vibration-driven cantilever to an underwater acoustic transducer with very low terminal voltage. The first stage reproduced the impedance framework of Liang and Liao in MATLAB/Simscape for standard energy harvesting (SEH), parallel synchronized switch harvesting on inductor (P-SSHI), and series synchronized switch harvesting on inductor (S-SSHI). Rather than comparing only waveform shape or one output-power value, the simulation extracted the fundamental components of the simulated piezoelectric voltage and equivalent current, reconstructed the complex electrical impedance, and compared harvested and extracted power with the analytical expressions. At the final vibration benchmark, the maximum simulated load powers were approximately 0.5 mW for SEH, 1.76 mW for S-SSHI, and 2.18 mW for P-SSHI.

For a representative P-SSHI run at $R_{\text{load}} = 1 \text{ M}\Omega$, the simulated harvested power was 2.09 mW, while Liang and Liao’s harvested-power expression predicted 2.10 mW. The simulated extracted power was 3.83 mW, compared with 3.71 mW from the paper’s analytical extracted-power expression. This agreement supported the use of the Simscape model as a basis for later circuit changes.

The ideal P-SSHI switch was then replaced with self-powered peak-detection circuitry based on the ESP-PSSHI topology of Zhang et al. The expected half-LC inversion time was 126.853 μs . The ideal switch closed for 124.916 μs with a measured delay of 40.525 μs from the equivalent-current zero crossing, while the BJT implementation closed for 133.849 μs but triggered 950.103 μs late. Its maximum harvested power was about 1.87 mW, showing that the duration of the LC inversion can be correct while the start time is still sufficiently delayed to reduce harvested power.

The same modeling workflow was then applied to an underwater acoustic equivalent source developed by Lutz Lampe. At 230 dB and 42 Hz, it gives an open-circuit amplitude of approximately 1.73 V. At 180 dB the same transducer representation gives only about

5.46 mV open circuit.

Once the circuit moved to LTspice, the goal changed from ideal switching to fast sweeps of real component SPICE models. A transformer/resonant matching network became particularly important. For a capacitive piezoelectric port, choosing the transformer primary magnetizing inductance near $L_p = 1/(\omega^2 C_p)$ cancels the piezoelectric susceptance at one operating frequency. At 42 Hz and 180 dB ($V_{oc} \approx 5.46$ mV), PSSHI had no measurable power in LTspice ($< 10^{-15}$ W). Then testing a transformer it reached approximately 32 nW with an MBR0520L diode rectifier and matched primary inductance with 0.5 winding ratio, whereas a realistic transformer candidate with series resistance produced only about 1.15 nW. At 230 dB ($V_{oc} \approx 1.73$ V), a low-loss, accurately tuned transformer model produced roughly 1.2 mW, while the compared P-SSHI model produced roughly 0.07 mW. However, small changes to L_p , C_p , or high winding resistance could reduce or reverse this advantage and finding a transformer with large inductances (40 H @ 42 Hz) is challenging. Thus, transformer matching can outperform SSHI at selected operating points, but it is much more sensitive to frequency, capacitance, primary inductance, and series resistance.

The overall result is not a single universally best interface. P-SSHI gave the largest power in the validated vibration benchmark, while carefully tuned transformer matching was stronger in millivolt level underwater simulation where the rectifier forward voltage cannot be reached. The practical design therefore depends on source voltage, source impedance, frequency stability, component loss, and phase consistency when multiple piezoelectric elements are combined.

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Chapter 1

Introduction

1.1 Motivation

Piezoelectric energy harvesters convert mechanical strain into electrical energy and can support low-power sensing systems where battery replacement is difficult. The useful output, however, depends on more than the piezoelectric material. The interface circuit determines when current is allowed to flow, how the piezoelectric capacitance is charged and discharged, how much energy is lost in switching and rectification, and the external load dependence.

This work began with a conventional vibration-driven benchmark because it provided published circuit equations, experimental parameters, and measured power curves against which a simulation could be checked. The same simulation framework was then extended to self-powered switching, underwater acoustic excitation, transformer matching, real component models, and multiple piezoelectric sources. These are successive simulations and circuit refinements directed at the same interface-design question.

1.2 Research Question

The research question used throughout the work is:

How can the electrical interface of a piezoelectric energy harvester be modeled and optimized so that useful power can be extracted under both conventional vibration excitation and extremely low-voltage underwater acoustic excitation?

The question has two connected parts. First, the model must be accurate enough that analytical equations and time-domain simulation agree for a known benchmark. Second, it

must demonstrate how to extract power at low source voltages, where ideal switches, fixed diode drops, and lossless magnetic elements are no longer valid assumptions.

1.3 Main Contributions

The main contributions of the work are:

1. a MATLAB/Simscape reproduction of the Liang–Liao SEH, P-SSHI, and S-SSHI impedance analysis, including direct comparison of waveforms, harvested power, and extracted power;
2. a timing-level and power analysis of a self-powered BJT P-SSHI detector;
3. application of equivalent underwater circuit representation at very low voltage levels;
4. a transition to LTspice for substantially faster component-level sweeps using real diode, transistor, inductor, and transformer models;
5. derivation and testing of a shunt-inductor/transformer impedance matching, including sensitivity to primary inductance, turns ratio, winding resistance, frequency, and piezo capacitance;
6. simulation of multiple piezoelectric elements with phase mismatch and alternative rectifier-combination strategies.
7. Choose circuit components to design and test a PCB.

Chapter 2

Analytical Model, MATLAB/Simscape Implementation, and Validation

2.1 Purpose and Validation Strategy

The first part of the work established a validated simulation framework before any underwater or component-level optimization was attempted. The vibration benchmark of Liang and Liao was useful because it provides the electromechanical equations, the SEH, P-SSHI, and S-SSHI interface equations, analytical expressions for equivalent impedance and power, and experimental power curves for comparison [1]. The objective here was not simply to reproduce those equations. The objective was to build the corresponding nonlinear interfaces in MATLAB/Simscape, extract the simulated waveforms, independently calculate the quantities used in the analytical model, and then determine whether the two approaches agreed.

2.2 Base-Excited Piezoelectric Model

2.2.1 Coupled electrical equations

Liang and Liao represent a base-excited piezoelectric harvester using a single-degree-of-freedom mechanical model coupled to the piezoelectric voltage v_p and current i_p [1]. The

mechanical equations may be mapped into an equivalent electrical series RLC branch,

$$v_{eq}(t) = -\frac{M}{\alpha_e} \ddot{y}(t), \quad (2.1)$$

$$i_{eq}(t) = \alpha_e \dot{x}(t), \quad (2.2)$$

$$L = \frac{M}{\alpha_e^2}, \quad (2.3)$$

$$R = \frac{D}{\alpha_e^2}, \quad (2.4)$$

$$C = \frac{\alpha_e^2}{K + K_p}. \quad (2.5)$$

The mechanical dynamics are therefore represented by a series L – R – C branch driven by v_{eq} , while C_p and the harvesting circuit form the electrical part. In the frequency domain,

$$V_{eq}(j\omega) = \left(j\omega L + R + \frac{1}{j\omega C} + Z_{elec}(j\omega) \right) I_{eq}(j\omega), \quad (2.6)$$

with

$$Z_{elec}(j\omega) = \frac{V_p(j\omega)}{I_{eq}(j\omega)}. \quad (2.7)$$

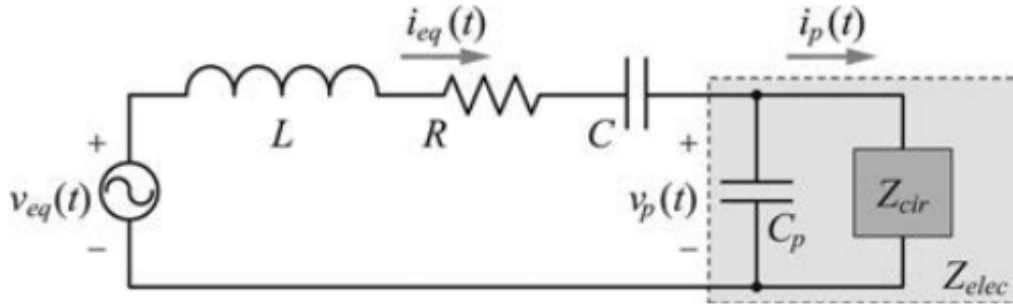


Figure 2.1: Base-excited piezoelectric circuit model used as the starting point for the Simscape model, reproduced from Liang and Liao [1].

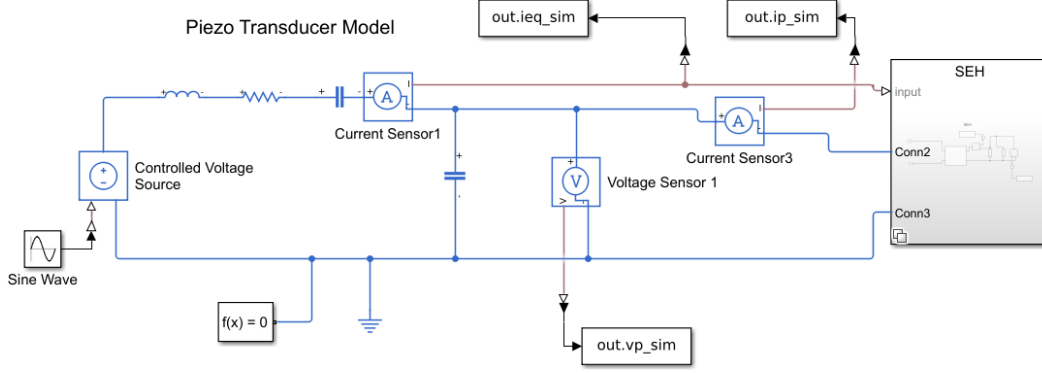


Figure 2.2: Simscape model. The interface circuit subsystem on the right can be set to SEH, P-SSHI or S-SSHI. Also contains voltage/current test points.

2.2.2 Benchmark parameters and source amplitude

The benchmark values used in the Simscape model were

Table 2.1: Vibration-benchmark parameters used for the MATLAB/Simscape reproduction.

Quantity	Symbol	Value
Excitation frequency	f	42 Hz
RMS base acceleration	$\ddot{Y}_{\text{rms}}$	10 m/s ²
Equivalent inductance	L	31 kH
Equivalent resistance	R	1 M Ω
Equivalent capacitance	C	448 pF
Piezoelectric capacitance	C_p	34.69 nF
Force-voltage factor	α_e	4.75×10^{-4}
Storage capacitor	C_{rect}	1 μ F
Bridge drop used in benchmark validation	V_F	1 V total
P-SSHI/S-SSHI inversion inductor	L_i	47 mH
Voltage inversion factor	γ	-0.7

These setting were used for all the data collection and analysis between the 3 interface circuits in this chapter for a controlled comparison.

The RMS acceleration was converted to peak acceleration,

$$\ddot{Y}_{pk} = \sqrt{2} \ddot{Y}_{\text{rms}}, \quad (2.8)$$

and because $M/\alpha_e = L\alpha_e$, the equivalent source amplitude used in the Simscape model was

$$V_{eq,pk} = L\alpha_e \ddot{Y}_{pk}. \quad (2.9)$$

For the values above this gives approximately 208 V. This large internal equivalent voltage should not be confused with the piezoelectric terminal voltage; it appears across the complete equivalent network.

The MATLAB implementation was

```
Y_rms_accel = 10;
y_peak_accel = sqrt(2) * Y_rms_accel;

L = 31e3;
R = 1e6;
C = 448e-12;
Cp = 34.69e-9;
ae = 4.75e-4;

VSrc_eq_amp = L * ae * y_peak_accel;
```

2.2.3 Open-circuit voltage check

Before connecting any interface, the open-circuit terminal voltage was checked from the equivalent circuit. With

$$Z_m = R + j\omega L + \frac{1}{j\omega C}, \quad (2.10)$$

$$Z_{C_p} = \frac{1}{j\omega C_p}, \quad (2.11)$$

the open-circuit terminal voltage is

$$V_{oc} = \left| V_{eq} \frac{Z_{C_p}}{Z_m + Z_{C_p}} \right|. \quad (2.12)$$

The corresponding code was

```
w = 2*pi*f;
Zseries_oc = R + 1j*w*L + 1/(1j*w*C);
ZCp_oc = 1/(1j*w*Cp);
```

```
openCircuitVoltage = abs( ...
    VSrc_eq_amp * ZCp_oc / (Zseries_oc + ZCp_oc) );
```

The simulated open-circuit waveform was also measured directly and its fundamental component was extracted. The measured fundamental amplitude used in the later validation was approximately 20 V. Liang and Liao's Eq. 44 is used to infer the equivalent source voltage from the measured open-circuit voltage which is used later in the harvested and extracted power calculations

$$V_{eq} = \frac{|R + j(X_L + X_C + X_{C_p})|}{|jX_{C_p}|} V_{oc}, \quad (2.13)$$

where $X_L = \omega L$, $X_C = -1/(\omega C)$, and $X_{C_p} = -1/(\omega C_p)$.

2.3 Simulated Waveforms Processing

2.3.1 Steady-state interval and final cycle

Each Simscape run was allowed to settle before any analytical comparison was made. A steady-state interval was retained for load-power averaging, and one complete final cycle was isolated for waveform, phase, impedance, and switching calculations. The current and voltage points which are used in the code below are measured as shown in Figure 2.2 and Figure 2.3.

```
idx_ss = t >= t_start_ss;
t_ss = t(idx_ss);
vp_ss = vp(idx_ss);
vstore_ss = vstore(idx_ss);
ieq_ss = ieq(idx_ss);

T = 1/f;
t2 = t_ss(end);
t1 = t2 - T;
idx_cycle = t >= t1 & t <= t2;

t_c = t(idx_cycle);
vp_c = vp(idx_cycle);
ieq_c = ieq(idx_cycle);
vstore_c = vstore(idx_cycle);
tau = t_c - t_c(1);
```

2.3.2 Fundamental extraction from the actual simulated signals

A key assumption in Liang and Liao is that the equivalent current can be regarded as a perfect sine wave,

$$i_{eq}(t) = I_0 \sin(\omega t), \quad (2.14)$$

and that only the fundamental harmonic of v_p materially affects the electromechanical dynamics [1]. In the simulation, however, neither the amplitude nor the phase was inserted from the analytical model. Both were obtained from the actual final-cycle waveforms which accounted for phase shifts.

Each simulated signal $y(t)$ is fitted to a single-harmonic model using the method of least squares:

$$y(t) \approx a \sin(\omega t) + b \cos(\omega t) \quad (2.15)$$

The fundamental amplitude and phase are then

$$A = \sqrt{a^2 + b^2}, \quad \phi = \text{atan2}(b, a), \quad (2.16)$$

and the reconstructed fundamental is

$$y_F(t) = A \sin(\omega t + \phi). \quad (2.17)$$

The exact calls used in the simulation were

```
[vp_F, VpF_amp, VpF_phase] = ...  
    fundamentalComponent(t_c, vp_c, f);  
  
[ieq_F_raw, I0_raw, ieq_phase_raw] = ...  
    fundamentalComponent(t_c, ieq_c, f);
```

and inside `fundamentalComponent`:

```
y = y - mean(y, "omitnan");  
w = 2*pi*f;  
  
A = [sin(w*t), cos(w*t)];  
coeff = A \ y;  
a = coeff(1);  
b = coeff(2);  
  
amp = hypot(a, b);
```

```

phase = atan2(b, a);
yF = amp * sin(w*t + phase);

```

The phasors were then formed from the fitted amplitudes and phases,

$$V_{p,F} = V_{p,F} e^{j\phi_v}, \quad (2.18)$$

$$I_{eq,F} = I_0 e^{j\phi_i}, \quad (2.19)$$

and the simulated equivalent impedance was

$$Z_{\text{elec,sim}} = \frac{V_{p,F}}{I_{eq,F}}. \quad (2.20)$$

This impedance is decomposed later and used to calculate the numerical power result. Thus, the analytical paper model was used as a validation target, while the quantities being compared against it came from the nonlinear Simscape waveforms themselves [9].

2.3.3 Rectified voltage and power calculated from simulation

The average storage voltage (the voltage across C_{rect}) was calculated over the retained steady-state interval,

$$\bar{V}_{store} = \frac{1}{t_b - t_a} \int_{t_a}^{t_b} v_{store}(t) dt, \quad (2.21)$$

and the sum of $\bar{V}_{store}$ and the forward voltage drop of the bridge rectifier V_f (twice the diode forward voltage) was used to get

$$V_{rect} = \bar{V}_{store} + V_F. \quad (2.22)$$

which is used in further equations. The implementation was

```

Vstore_avg = trapz(t_ss, vstore_ss) / ...
              (t_ss(end) - t_ss(1));

Vrect_sim = Vstore_avg + VF_bridge;

```

Two power quantities were calculated independently.

First, useful load power was evaluated from the full steady state storage-voltage

waveform rather than from one sample:

$$P_{load} = \left\langle \frac{v_{store}^2(t)}{R_{load}} \right\rangle, \quad (2.23)$$

The code was

```
Pload_inst = vstore_ss.^2 ./ Rload;
Pload_avg = trapz(t_ss, Pload_inst) / ...
    (t_ss(end) - t_ss(1));
```

Second, total electrical power extracted from the equivalent source was calculated directly over the final cycle,

$$P_{\Delta, time} = \langle v_p(t) i_{eq}(t) \rangle, \quad (2.24)$$

using

```
Pdelta_time = trapz(t_c, vp_c .* ieq_c_eff) / ...
    (t_c(end) - t_c(1));
```

The comparison with paper equations and simulation was made after extracting I_0 , phase, and V_{rect} from the simulation itself and then evaluated all the paper equations including piezo voltage, impedance and power.

2.4 Standard Energy Harvesting (SEH)

2.4.1 SEH model

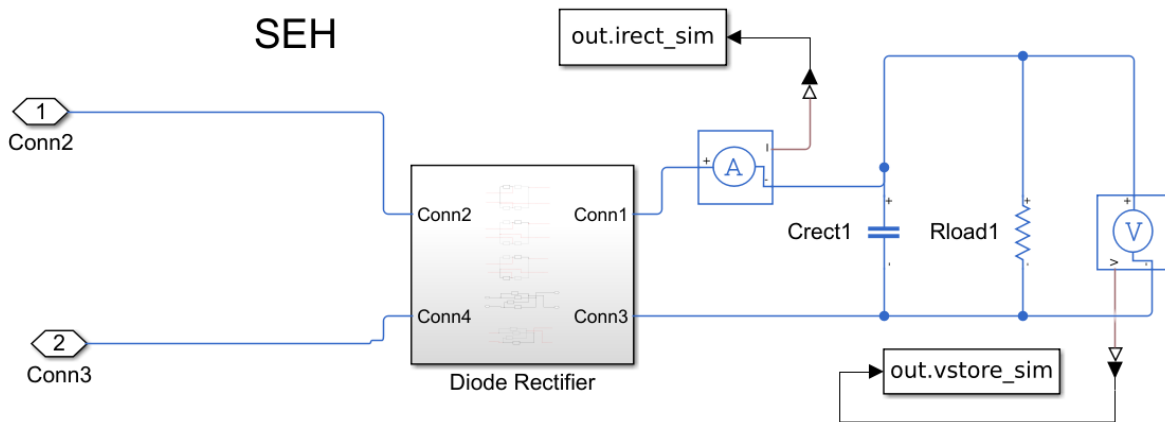


Figure 2.3: Internals of Simscape subsystem on the right from Figure 2.2 for SEH interface circuit containing rectifier and smoothing capacitor C_{rect} with R_{load} .

2.4.2 SEH waveform and impedance equations

For SEH, the full-wave bridge clamps v_p to $\pm V_{rect}$ when the rectifier conducts. During the blocked interval, the equivalent current charges or discharges C_p . With the sine-wave equivalent current of Eq. (2.14), the open-circuit paper voltage magnitude is

$$V_{ocp} = \frac{I_0}{\omega C_p}. \quad (2.25)$$

The I_0 in the above equation is with the interface circuit connected, so it is not be confused with the actual open circuit voltage V_{oc} . TODO: replace all the paper Voc with Vocp

Define the normalized rectified voltage

$$\tilde{V}_{rect} = \frac{V_{rect}}{V_{ocp}}. \quad (2.26)$$

The rectifier blocked angle θ is related to V_{rect} by

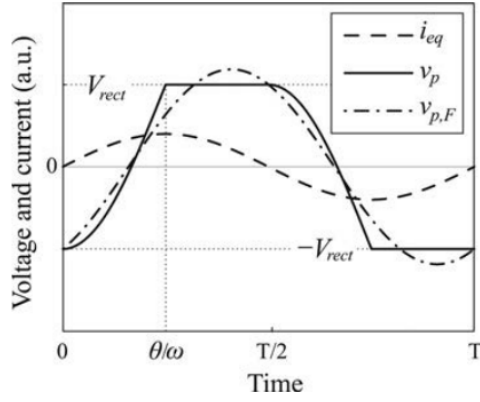
$$\cos \theta = 1 - 2\tilde{V}_{rect}. \quad (2.27)$$

The paper's one-cycle SEH voltage waveform is then

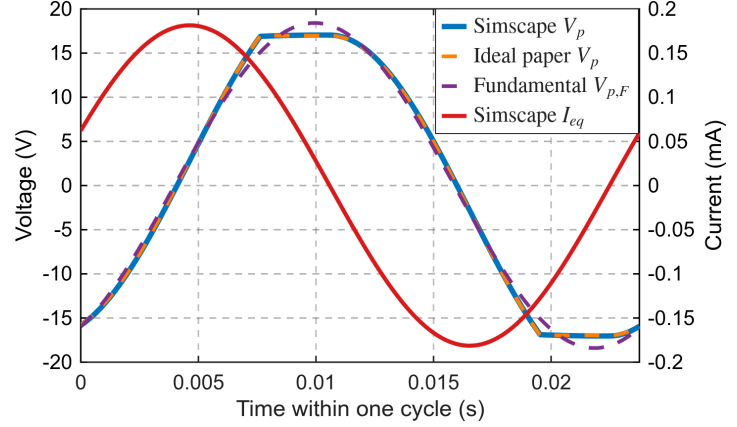
$$v_p(t) = \begin{cases} V_{oc}[1 - \cos(\omega t)] - V_{rect}, & 0 \leq \omega t < \theta, \\ V_{rect}, & \theta \leq \omega t < \pi, \\ V_{rect} - V_{oc}[1 + \cos(\omega t)], & \pi \leq \omega t < \pi + \theta, \\ -V_{rect}, & \pi + \theta \leq \omega t < 2\pi. \end{cases} \quad (2.28)$$

The Code Snippet: 2.1 shows how the above waveform was generated.

As shown in the graph below, this allowed for a direct overlay of the simulated $v_p(t)$, the paper waveform, its fitted fundamental, and the simulated equivalent current on the same final cycle.



(a) Paper theoretical curves.



(b) Comparison between Simscape simulated SEH V_p curve and ideal paper Eq. (2.28) V_p .

Figure 2.4: Simulation results look visually identical to paper graph. The measured solid-blue V_p from the simulation is directly matched with the dashed-orange paper V_p , Eq. (2.28). The red I_{eq} curve is also out of phase like in the paper graph. Figure (a) taken from [1, Fig. 3].

The fundamental component obtained analytically from this Eq. (2.28) waveform is

$$v_{p,F}(t) = \frac{I_0}{2\pi\omega C_p} ([\sin(2\theta) - 2\theta] \cos(\omega t) + 2 \sin^2 \theta \sin(\omega t)), \quad (2.29)$$

and therefore

$$Z_{\text{elec,SEH}} = \frac{1}{\pi\omega C_p} [\sin^2 \theta + j(\sin \theta \cos \theta - \theta)]. \quad (2.30)$$

To distinguish useful harvested power from circuit loss, the real part of Z_{elec} is decomposed into a harvesting resistance R_h and a dissipative resistance R_d , while the imaginary part is represented by X_E . With

$$\tilde{V}_F = \frac{V_F}{V_{oc}}, \quad (2.31)$$

Liang and Liao give for SEH

$$R_d = \frac{4}{\pi\omega C_p} \tilde{V}_F (1 - \tilde{V}_{rect}), \quad (2.32)$$

$$R_h = \frac{4}{\pi\omega C_p} (\tilde{V}_{rect} - \tilde{V}_F) (1 - \tilde{V}_{rect}), \quad (2.33)$$

$$X_E = \frac{1}{\pi\omega C_p} (\sin \theta \cos \theta - \theta). \quad (2.34)$$

Therefore,

$$Z_{\text{elec}} = (R_d + R_h) + jX_E. \quad (2.35)$$

2.4.3 SEH power and load sweep

After obtaining R_h , R_d , and X_E , the analytical harvested and extracted powers were calculated with Liang and Liao's Eqs. 42 and 43:

$$P_h = \frac{V_{eq}^2}{2} \frac{R_h}{(X_L + X_C + X_E)^2 + (R + R_d + R_h)^2}, \quad (2.36)$$

$$P_\Delta = \frac{V_{eq}^2}{2} \frac{R_h + R_d}{(X_L + X_C + X_E)^2 + (R + R_d + R_h)^2}. \quad (2.37)$$

At a representative $1 \text{ M}\Omega$ load point, the result was approximately

Table 2.2: Representative SEH power comparison at $R_{load} = 1 \text{ M}\Omega$.

Quantity	Power
Simscape harvested/load power, Eq. (2.23)	0.255 mW
Calculated harvested power, Eq. (2.36)	0.261 mW
Simscape extracted power, Eq. (2.24)	0.278 mW
Calculated extracted power, Eq. (2.37)	0.278 mW

Running a load sweep from 10^4 to 10^7 , the result was approximately

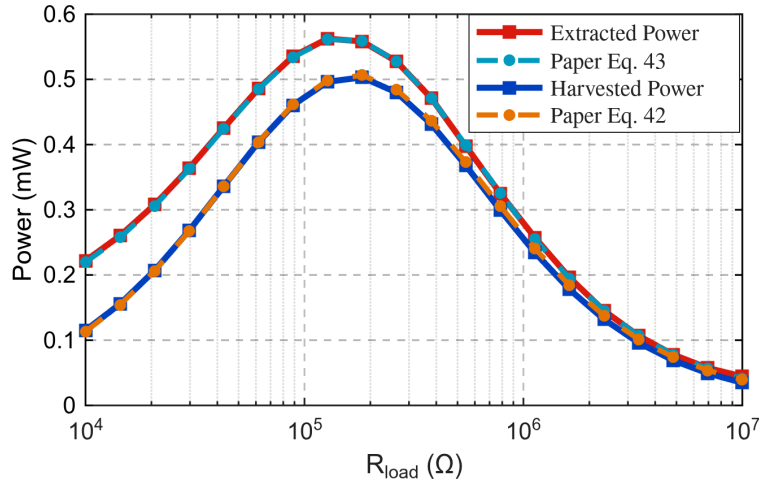


Figure 2.5: SEH R_{load} sweep. The simulated harvested-power and Eq. 42 curves are closely matched, same with the extracted-power and Eq. 43 curves. The extracted power shows the larger total energy extracted from the system.

The SEH sweep gave

$$R_{load,opt} \approx 183 \text{ k}\Omega, \quad P_{load,max} \approx 0.5 \text{ mW}. \quad (2.38)$$

2.5 Parallel SSHI (P-SSHI)

2.5.1 Ideal P-SSHI switching model

P-SSHI adds a switched inductor directly in parallel with the piezoelectric element. The switch is triggered at the displacement extremum, corresponding to the equivalent-current zero crossing in the ideal harmonic model. During the brief switching interval, L_i and C_p form an LC resonant loop and invert the piezoelectric voltage before the next half-cycle.

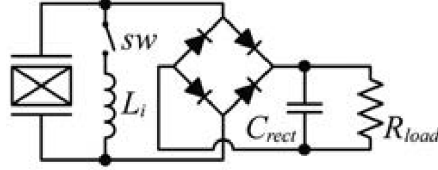


Figure 2.6: P-SSHI interface circuit topology. Figure taken from [1, Fig. 4].

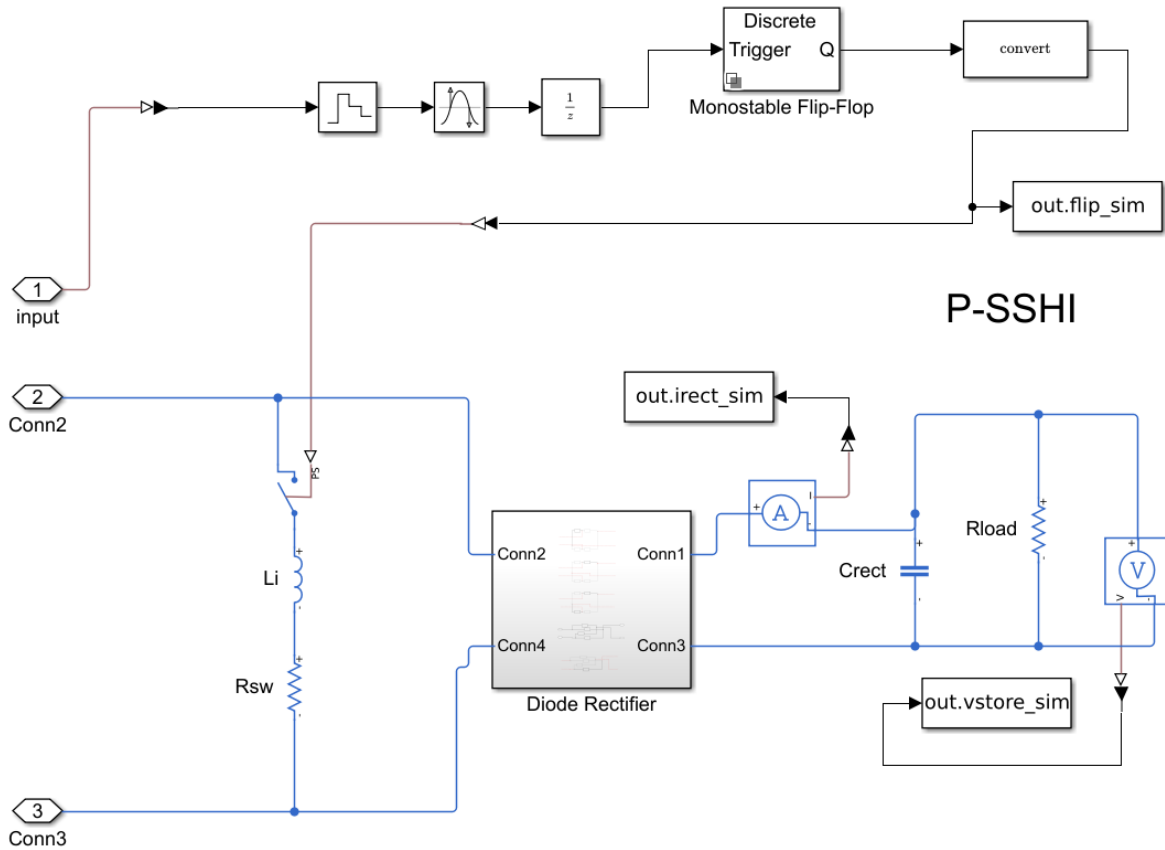


Figure 2.7: Internals of Simscape subsystem on the right from Figure 2.2 for P-SSHI interface circuit. Contains equivalent-current zero cross detector triggered for fixed duration T_{sw} (ideal non-circuit implementation) with ideal switch and inductor L_i / resistor R_{sw} , followed by rectifier and smoothing capacitor C_{rect} with R_{load} .

The half-LC switching interval is

$$T_{sw} = \pi \sqrt{L_i C_p}, \quad (2.39)$$

which gives approximately 127 μs for $L_i = 47 \text{ mH}$ and the benchmark C_p . This is much shorter than the approximately 23.8 ms mechanical period at 42 Hz, so the analytical model treats the inversion as effectively instantaneous.

The voltage inversion factor gamma used in the paper is related to the switching-loop quality factor by

$$\gamma = -\exp\left(-\frac{\pi}{2Q}\right). \quad (2.40)$$

where the quality factor Q (r is the resistance in the C_p - L_i - r loop including R_{sw} , and L_i/C_p series resistance) is:

$$Q = \frac{1}{r} \sqrt{\frac{L_i}{C_p}}. \quad (2.41)$$

The benchmark simulations used $\gamma = -0.7$.

2.5.2 P-SSHI waveform and impedance equations

The P-SSHI one-cycle piezoelectric voltage waveform is

$$v_p(t) = \begin{cases} V_{oc}[1 - \cos(\omega t)] - \gamma V_{rect}, & 0^+ \leq \omega t < \theta, \\ V_{rect}, & \theta \leq \omega t \leq \pi^-, \\ \gamma V_{rect} - V_{oc}[1 + \cos(\omega t)], & \pi^+ \leq \omega t < \pi + \theta, \\ -V_{rect}, & \pi + \theta \leq \omega t \leq 2\pi^-. \end{cases} \quad (2.42)$$

The blocked angle now satisfies

$$\cos \theta = 1 - (1 + \gamma) \tilde{V}_{rect}. \quad (2.43)$$

The Code Snippet: 2.1 shows how the above waveform was generated.

The resulting analytical waveform was overlaid with the simulated v_p . This comparison is more restrictive than comparing only power because it checks the rectifier-blocked interval, clamped interval, inversion event, and phase relative to i_{eq} .

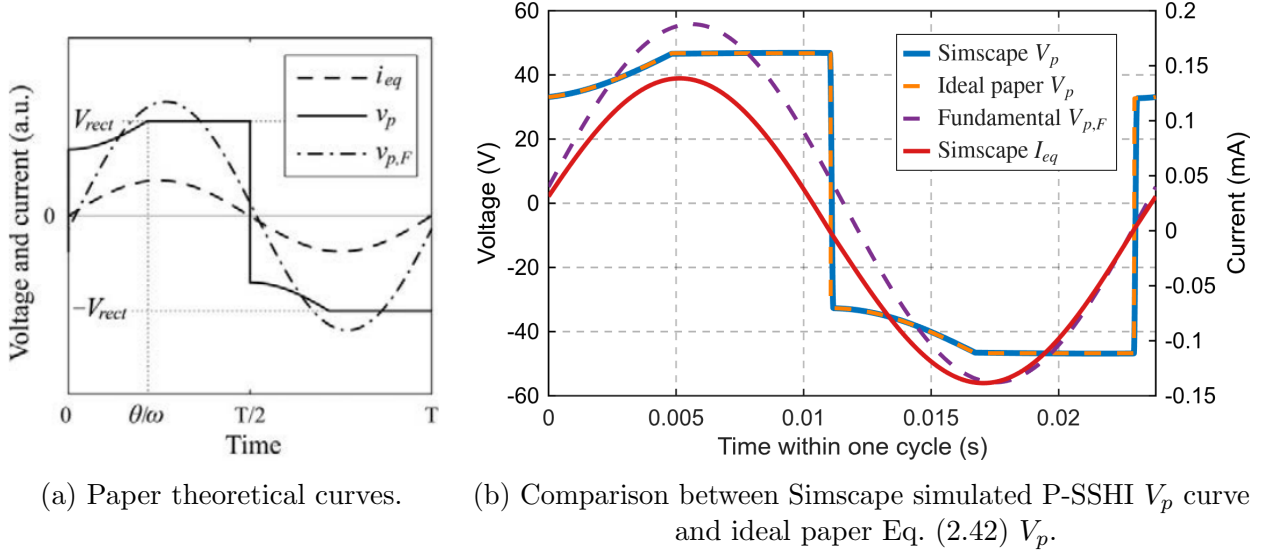


Figure 2.8: Simulation results look visually identical to paper graph. The measured solid-blue V_p from the simulation is directly matched with the dashed-orange paper V_p , Eq. (2.42). The red I_{eq} curve is also now in phase like in the paper graph. Figure (a) taken from [1, Fig. 5].

The corresponding electrical equivalent impedance is

$$Z_{\text{elec,PSSHI}} = \frac{1}{\pi\omega C_p} \left\{ (1 - \cos\theta) \left[\frac{4}{1 + \gamma} - 1 + \cos\theta \right] + j(\sin\theta \cos\theta - \theta) \right\}. \quad (2.44)$$

The P-SSHI impedance components used in the power calculation are

$$R_d = \frac{1}{\pi\omega C_p} \left\{ 2\tilde{V}_F[2 - \tilde{V}_{rect}(1 + \gamma)] + \tilde{V}_{rect}^2(1 - \gamma^2) \right\}, \quad (2.45)$$

$$R_h = \frac{2}{\pi\omega C_p} (\tilde{V}_{rect} - \tilde{V}_F)[2 - \tilde{V}_{rect}(1 + \gamma)], \quad (2.46)$$

$$X_E = \frac{1}{\pi\omega C_p} (\sin\theta \cos\theta - \theta). \quad (2.47)$$

These values were then substituted into Eqs. (2.36) and (2.37) without changing the power equations themselves.

2.5.3 P-SSHI power and load sweep

At a representative $1\text{ M}\Omega$ load point, the result was approximately

Table 2.3: Representative P-SSHI power comparison at $R_{load} = 1 \text{ M}\Omega$.

Quantity	Power
Simscape harvested/load power, Eq. (2.23)	2.09 mW
Calculated harvested power, Eq. (2.36)	2.10 mW
Simscape extracted power, Eq. (2.24)	3.83 mW
Calculated extracted power, Eq. (2.37)	3.71 mW

The simulated harvested power rises as the output operating point becomes more favorable, reaches a maximum, and then falls as the increasingly high load draws too little useful dc current. In contrast, the total extracted power can continue increasing over part of this range because a larger portion of the extracted energy is not necessarily delivered to the load. TODO: why extracted increase, but not in SEH

Running a load sweep from 10^4 to 10^7 , the result was approximately

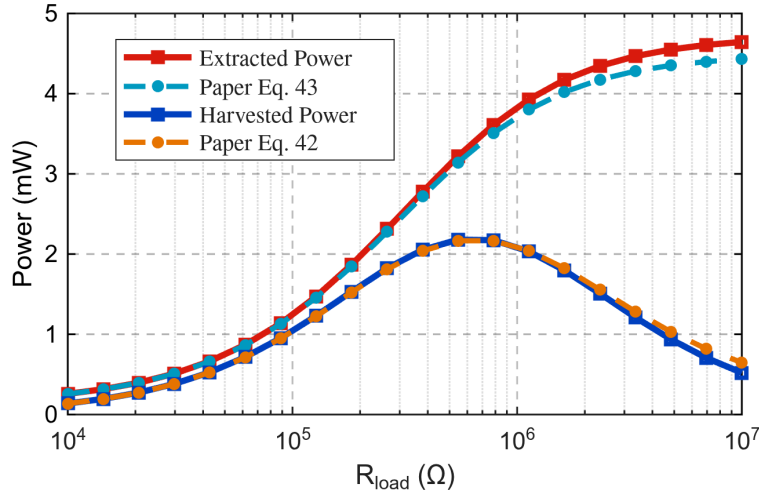


Figure 2.9: P-SSHI R_{load} sweep. The simulated power and equation curves are still closely matched. The useful harvested-power maximum is much higher than SEH.

The P-SSHI sweep gave

$$R_{load,opt} \approx 546 \text{ k}\Omega, \quad P_{load,max} \approx 2.18 \text{ mW}. \quad (2.48)$$

2.6 Series SSHI (S-SSHI)

2.6.1 Ideal S-SSHI switching model

S-SSHI places the switched inductor in series with the rectifier path rather than directly across the piezoelectric element. The same short inversion-time condition in Eq. (2.39) applies, but the resulting ideal voltage waveform differs from P-SSHI.

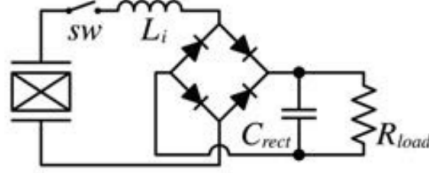


Figure 2.10: S-SSHI interface circuit topology. Figure taken from [1, Fig. 4].

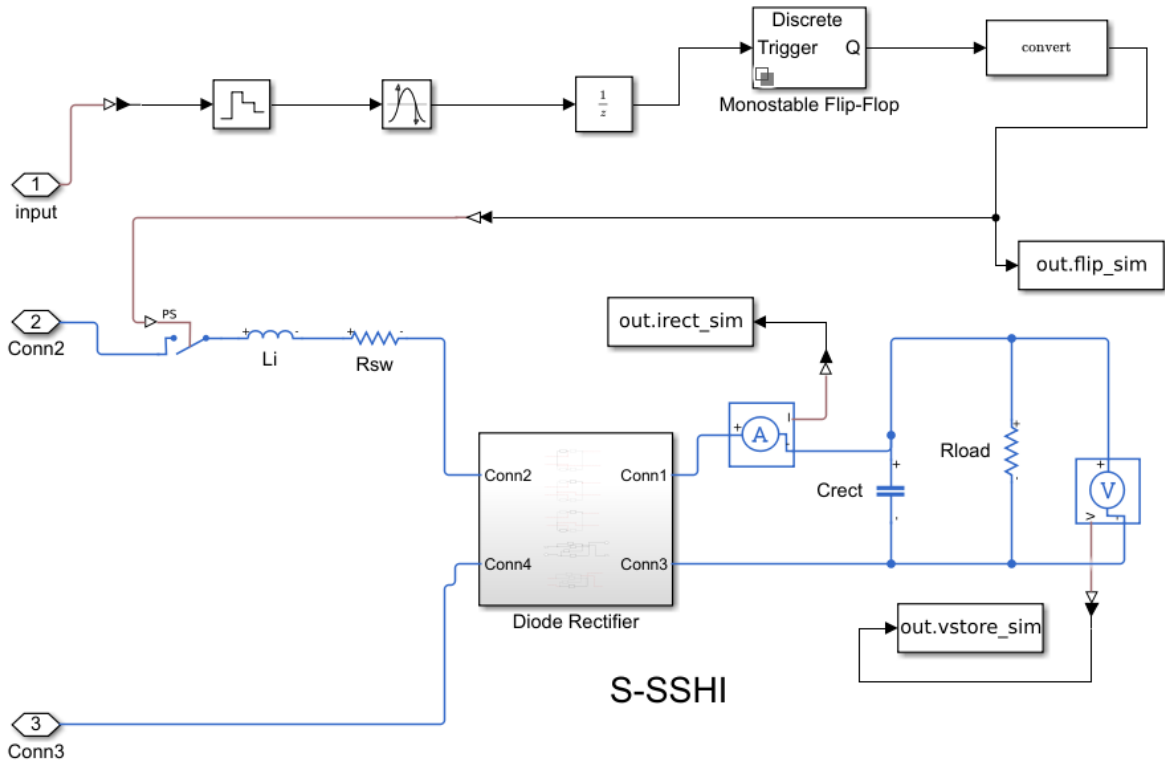


Figure 2.11: Internals of Simscape subsystem on the right from Figure 2.2 for S-SSHI interface circuit. Contains equivalent-current zero cross detector triggered for fixed duration T_{sw} (ideal non-circuit implementation) with ideal switch and inductor L_i / resistor R_{sw} , followed by rectifier and smoothing capacitor C_{rect} with R_{load} .

2.6.2 S-SSHI waveform and impedance equations

Liang and Liao give

$$v_p(t) = \begin{cases} -V_{oc} \cos(\omega t) + \frac{1-\gamma}{1+\gamma}(V_{oc} - V_{rect}), & 0^+ \leq \omega t \leq \pi^-, \\ -V_{oc} \cos(\omega t) - \frac{1-\gamma}{1+\gamma}(V_{oc} - V_{rect}), & \pi^+ \leq \omega t \leq 2\pi^-. \end{cases} \quad (2.49)$$

Unlike SEH and P-SSHI, S-SSHI does not use the same rectifier blocked angle θ in the analytical waveform.

The Code Snippet: 2.1 shows how the above waveform was generated.

As with SEH and P-SSHI, this paper waveform was compared against the actual Simscape $v_p(t)$.

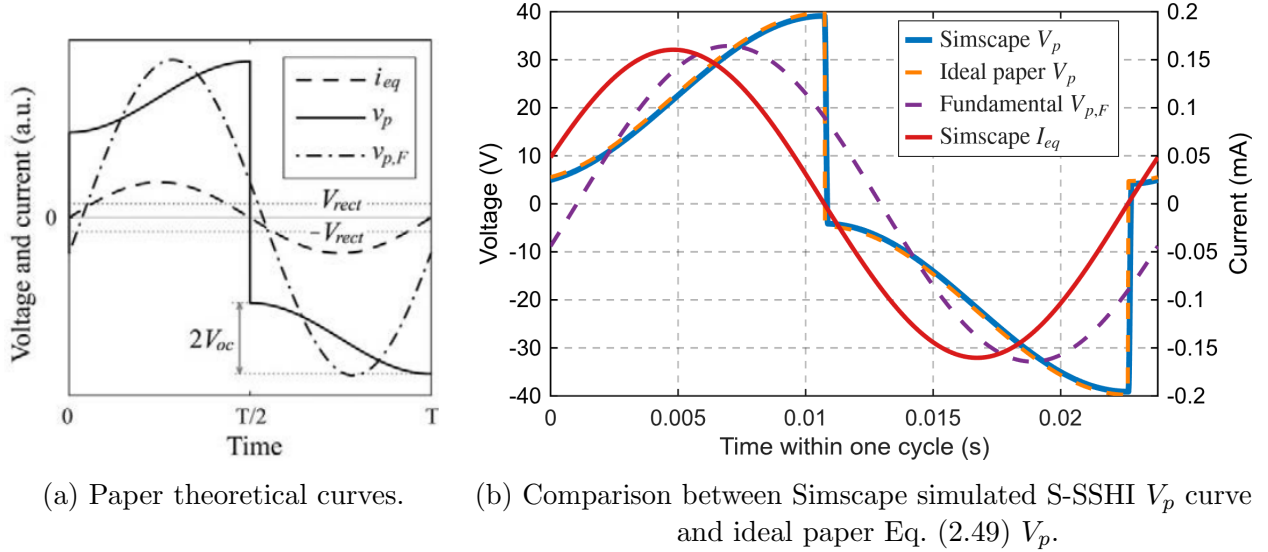


Figure 2.12: Simulation results look visually identical to paper graph. The measured solid-blue V_p from the simulation is directly matched with the dashed-orange paper V_p , Eq. (2.49). The red I_{eq} curve is also in phase like in the paper graph. Figure (a) taken from [1, Fig. 5].

Its equivalent impedance is

$$Z_{elec,SSHI} = \frac{1}{\omega C_p} \left[\frac{4}{\pi} \frac{1-\gamma}{1+\gamma} (1 - \tilde{V}_{rect}) - j \right]. \quad (2.50)$$

The impedance components used in Eqs. (2.36) and (2.37) are

$$R_d = \frac{4}{\pi\omega C_p} \frac{1-\gamma}{1+\gamma} (1 - \tilde{V}_{rect} + \tilde{V}_F)(1 - \tilde{V}_{rect}), \quad (2.51)$$

$$R_h = \frac{4}{\pi\omega C_p} \frac{1-\gamma}{1+\gamma} (\tilde{V}_{rect} - \tilde{V}_F)(1 - \tilde{V}_{rect}), \quad (2.52)$$

$$X_E = -\frac{1}{\omega C_p}. \quad (2.53)$$

TODO: add commentray on voltage/current levels compared to SEH, also add in SEH/PSSHI secitons

2.6.3 S-SSHI power and load sweep

At a representative $0.1 \text{ M}\Omega$ load point, the result was approximately

Table 2.4: Representative S-SSHI power comparison at $R_{load} = 0.1 \text{ M}\Omega$.

Quantity	Power
Simscape harvested/load power, Eq. (2.23)	1.59 mW
Calculated harvested power, Eq. (2.36)	1.6 mW
Simscape extracted power, Eq. (2.24)	2.21 mW
Calculated extracted power, Eq. (2.37)	2.23 mW

Running a load sweep from 10^4 to 10^7 , the result was approximately

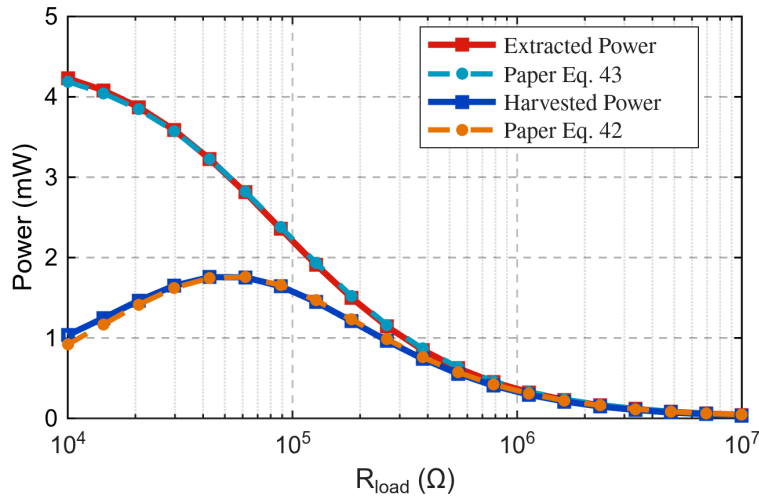


Figure 2.13: S-SSHI R_{load} sweep. The waveforms are still closely matched. The useful harvested-power maximum is a bit lower than P-SSHI but substantially higher than SEH.

The maximum occurred at a much smaller load resistance than the P-SSHI maximum. The S-SSHI sweep gave

$$R_{load,opt} \approx 43 \text{ k}\Omega, \quad P_{load,max} \approx 1.76 \text{ mW}. \quad (2.54)$$

2.7 Overall Interface Circuit Comparison

Liang and Liao's compares theoretical and experimental harvested power for SEH, P-SSHI, and S-SSHI. Their P-SSHI case produced the greatest harvestable power, followed by S-SSHI and then SEH at around 2.5 mW, 2.0 mW and 0.5 mW respectively [1].

The Simscape load sweeps reproduced the same order at slightly less power for the SSHI cases which may be due to the small difference in the exact gamma value in Eq. (2.40) and corresponding parasitic resistance. The parasitic resistance (inductor series resistance + other r in the loop) may be calculated slightly differently in the paper and a slight difference can have a big impact in the harvested power. The final Simscape load-sweep maxima were

Table 2.5: Final vibration-benchmark maxima from the Simscape load sweeps.

Interface	Optimum R_{load}	Maximum load power	Relative to SEH
SEH	183 k Ω	0.5 mW	1.0 $\times$
S-SSHI	43 k Ω	1.76 mW	3.6 $\times$
P-SSHI	546 k Ω	2.18 mW	4.4 $\times$

Thus, ideal P-SSHI produced the largest harvested power for the vibration benchmark. In SEH a lot of piezo current is spent charging and discharging the piezo capacitance. First it charges C_p and conducts, but at the end of the half cycle the current reverses, but the voltage does not reverse right away, so generated charge is wasted discharging/ recharging C_p . In PSSHI instead of slowly charging/discharging, the inductor-switch branch flips the piezo voltage before the next half-cycle to start with the negative opposite voltage instead of slowly charging. So it removes the dead-zone by switching the AC voltage instantly, and v_p and i_{eq} stay in phase.

More importantly for the remainder of the work, the close waveform, impedance, power, and load-sweep agreement showed that the Simscape model was sufficiently consistent with the published analytical framework to use as the baseline for replacing the ideal switching action with real switching circuitry and, later, for changing the source to the underwater equivalent model.

2.8 Paper Waveforms and Load-Sweep Processing

2.8.1 Paper Waveforms Generation

In all three cases, the numerical calculations needed I_0 , phase, and V_{rect} from the simulation itself. The code first used the fitted I_0 to obtain V_{oc} , then used the simulated V_{rect} to calculate $\tilde{V}_{rect}$ and θ , and finally evaluated each V_p equation point by point as in Eq. (2.28), Eq. (2.42) and Eq. (2.49) while also accounting for the I_{eq} phase shift.

P-SSHI and S-SSHI both use the same SSHI inversion factor gamma. SEH is the gamma = 1 special case of the P-SSHI formulas.

Listing 2.1: MATLAB Code for Paper Equations and Waveform Generation

```
Qsshi = sqrt(Li/Cp) / Rloop_total;
gamma_sshi = -exp(-pi/(2*Qsshi));
if Type == "SEH"
    gamma = 1;
elseif Type == "PSSHI" || Type == "SSSHI"
    gamma = gamma_sshi;

Voc_sim = I0 / (w*Cp);
Vrect_sim = Vstore_avg + VF_bridge;
Vtilde_sim = Vrect_sim / Voc_sim;
theta_paper = acos(1 - (1 + gamma)*Vtilde_sim);

phi = mod(w*t_c + ieq_phase, 2*pi);
vp_paper = nan(size(phi));

if Type == "SSSHI"
    offset = ((1 - gamma)/(1 + gamma)) * (Voc_sim - Vrect_sim);

    for kk = 1:length(phi)
        p = phi(kk);
        if p >= 0 && p < pi
            vp_paper(kk) = -Voc_sim*cos(p) + offset;
        else
            vp_paper(kk) = -Voc_sim*cos(p) - offset;
        end
    end
end
```

```

elseif Type == "SEH" || Type == "PSSHI"

    for kk = 1:length(phi)
        p = phi(kk);

        % General SEH/P-SSHI Eq. 21. SEH is the special case gamma = 1.
        if p >= 0 && p < theta_paper
            vp_paper(kk) = Voc_sim * (1 - cos(p)) - gamma * Vrect_sim;
        elseif p >= theta_paper && p < pi
            vp_paper(kk) = Vrect_sim;
        elseif p >= pi && p < pi + theta_paper
            vp_paper(kk) = gamma * Vrect_sim - Voc_sim * (1 + cos(p));
        else
            vp_paper(kk) = -Vrect_sim;
        end
    end
end
end

```

2.8.2 Load-Resistance Sweep Method

The load sweep was performed by changing R_{load} while changing the simulation duration. The output time constant changes as

$$\tau_{RC} = R_{load}C_{rect}, \quad (2.55)$$

so high-resistance loads require a longer settling time.

The script therefore generated a logarithmic load vector and adjusted the simulation stop time according to the output time constant:

```

Rload_list = logspace(4, 7, 20);

for k = 1:numel(Rload_list)
    Rload_k = Rload_list(k);
    tauRC = Rload_k * Crect;

    stopTime_k = max(minStopTime, tauMultiplier*tauRC);
    stopTime_k = min(stopTime_k, maxStopTime);
end

```

```

    t_start_ss_k = ssStartFraction * stopTime_k;

    sweepResults(k) = runOnePEH( ...
        ..., Rload_k, ..., ...
        stopTime_k, t_start_ss_k, ...);
end

```

For each load value the same steady-state and fundamental extraction described above was repeated. The optimum was selected using the simulated average dc load power:

```

validP = isfinite(sweepTable.Pload_avg_W);

[Pmax, idxLocal] = ...
    max(sweepTable.Pload_avg_W(validP));

validIdx = find(validP);
idxOpt = validIdx(idxLocal);

Rload_opt = sweepTable.Rload_ohm(idxOpt);
Pload_opt = sweepTable.Pload_avg_W(idxOpt);

```

TODO: Use actual equations, Eq. (2.36) and Eq. (2.37) instead of saying eq 42, eq 43 without reference.

The sweep stored, for every R_{load} , the simulated harvested power, time-domain extracted power, analytical Eq. 42 power, analytical Eq. 43 power, V_{store} , V_{rect} , $\tilde{V}_{rect}$, $Z_{elec,sim}$, $Z_{elec,paper}$, R_d , R_h , and X_E . This made the load sweep a full model-to-model comparison rather than only an output-power search.

Chapter 3

Self-Powered and Alternative SSHI Switching

3.1 Self-Powered P-SSHI Circuit

Zhang et al. proposed an efficient self-powered P-SSHI interface in which passive peak-detection branches trigger the inductor path without an externally powered timing source [2]. The published circuit uses separate positive and negative detection paths so that the piezoelectric capacitance and an inductor form the inversion loop after the detected voltage peak.

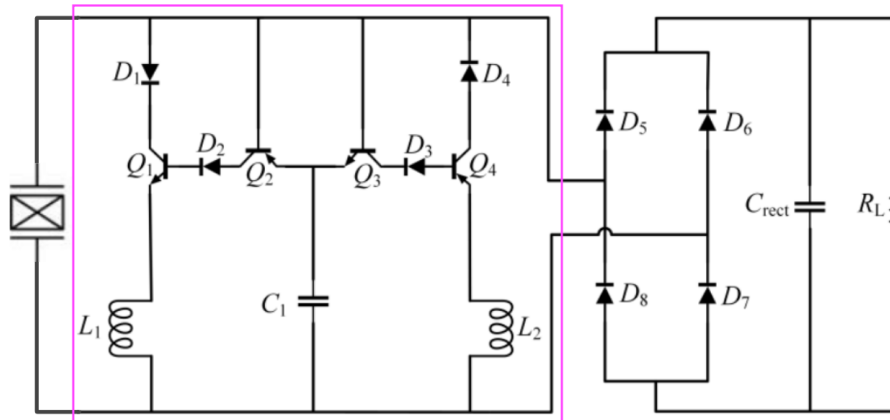


Figure 3.1: ESP-PSSHI circuit used as the self-powered switching implementation. The ideal switch and inductor is replaced with the outlined area. Figure taken from [2, Fig. 6]

The Enhanced Self-Powered Parallel Synchronous Switch Harvesting on Inductor (ESP-PSSHI) circuit eliminates the need for external digital controllers, active sensors, or external

power supplies by using a passive peak-detector and envelope-tracking configuration to trigger voltage inversion.

3.1.1 Circuit Operation and Mathematical Formulation

1. First v_p rises to the positive peak.
2. C_1 charges and stores/holds that peak.
3. v_p starts falling after the current zero-crossing.
4. C_1 is still high, so $v_{C1} - v_p$ grows.
5. When $v_{C1} - v_p > v_{be}$, Q_2 turns on.
6. Q_2 drives Q_1 on.
7. Once Q_1 is on, the piezo capacitance C_p has a path through $D_1 \rightarrow Q_1 \rightarrow L_1 \rightarrow C_p$.
8. The detection capacitor C_1 also has a path through $Q_2 \rightarrow D_2 \rightarrow Q_1 \rightarrow L_1 \rightarrow C_1$.
9. C_p discharges into L_1 and v_p flips polarity, C_1 also discharges.
10. At the end of that half-cycle, L_1 current falls to zero. D_1 doesn't conduct, Q_2 turns off, and the branch opens.
11. An identical network (D_3, Q_3, Q_4, D_4, L_2) handles the negative half-cycle.

The switching action happens as the transistor conducts when the voltage difference between the peak-hold capacitor C_1 and the decreasing piezoelectric voltage v_p exceeds the required semiconductor threshold voltage:

$$v_{C1} - v_p \geq V_{BE} \quad (3.1)$$

where V_{BE} is the base-emitter turn-on voltage of transistor Q_2 . So v_p needs to slightly drop before the switching action happens which causes a slight inefficiency.

3.2 Measured Trigger Delay and Inversion Duration

The important part in this simulation was *how long* the switch remained on and *when* it began to conduct. From Eq. (2.39), using $L_i = 47$ mH and the benchmark C_p , the expected

ideal half-LC interval was

$$T_{sw} = \pi \sqrt{L_i C_p} = 126.853 \mu\text{s}. \quad (3.2)$$

The timing measured from the final steady-state cycle was:

Table 3.1: Measured switching timing in the benchmark P-SSHI simulation at $R_{load} = 1 \text{ M}\Omega$.

Switch implementation	Closed duration	Delay from i_{eq} zero crossing
Ideal/control switch	125 μs	16 μs
BJT self-powered detector	134 μs	952 μs

The BJT duration was close to the expected LC interval, but the branch began nearly 1 ms after the desired event. So it does not fully recover the ideal P-SSHI result.

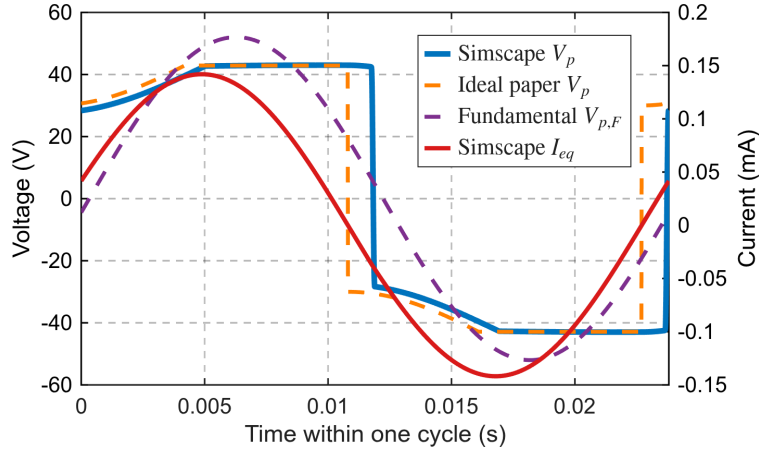


Figure 3.2: The Simscape simulated ESP-PSSHI V_p curve and ideal paper Eq. (2.42) V_p . The ideal V_p switches at the current zero-crossing, but the Simscape V_p closes nearly 1 ms later.

At the tested load sweep, the self-powered BJT P-SSHI maximum was about 1.87 mW, compared with about 2.18 mW for the ideal-switch P-SSHI configuration at the same sweep settings. The BJT circuit therefore retained most of the SSHI benefit, but its delay was a measurable source of loss.

3.3 Passive Detector Components

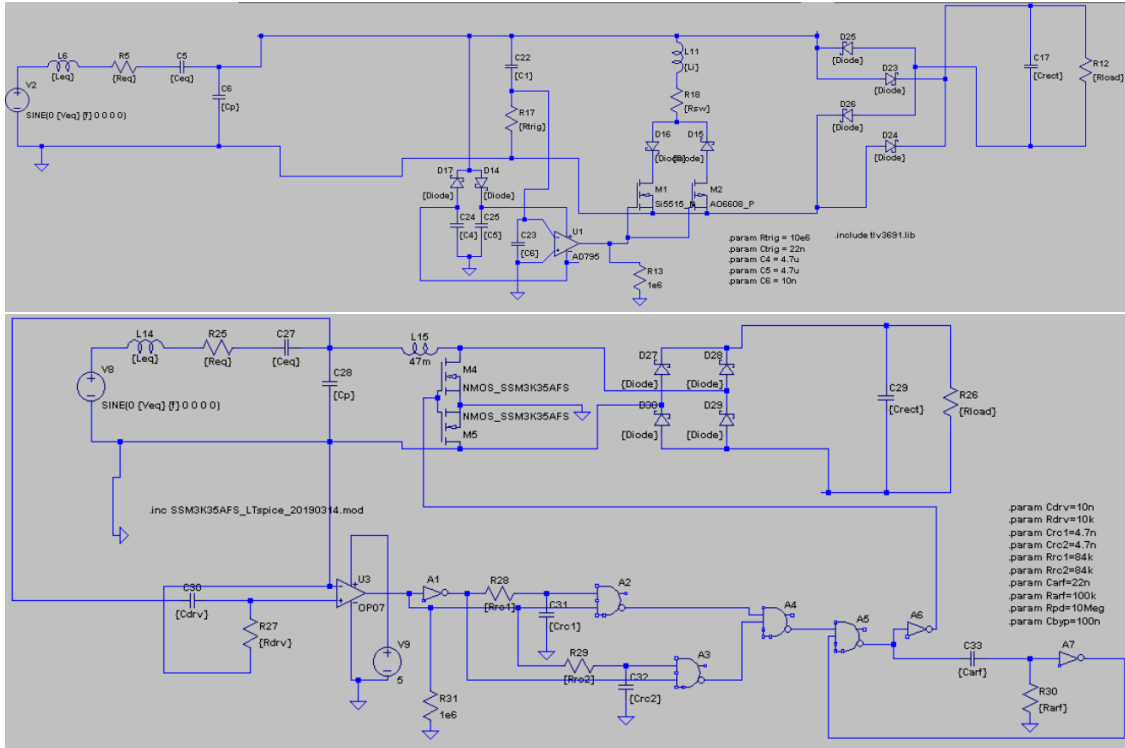
This design added a detection capacitor, NPN/PNP transistors and extra diodes. The table below shows the values and models used.

Table 3.2: Vibration-benchmark parameters used for the MATLAB/Simscope reproduction.

Quantity	Symbol	Value/Model
Peak detector capacitance	C_1	2.2 nF
PNP Transistors	Q_2, Q_4	TIP32C
NPN Transistors	Q_1, Q_3	TIP31C
Diodes	$D_1 - D_8$	BAT54

3.4 Two Active Zero-Cross / Peak-Detection Variants

Two additional active timing circuits were implemented in LTspice to test whether a powered detector could trigger the inversion more accurately.



The likely limitation was timing rather than simply lack of active circuitry. The tested active versions detected a voltage derivative or voltage peak, whereas the desired inversion start in the simulated P-SSHI waveform was tied to the zero crossing of the equivalent current. Related literature confirms that switching-time generation is a central SSHI design problem. Zouari et al. compared op-amp/MOSFET and bipolar self-powered switching techniques and reported much higher simulated power from the bipolar approach after loss reduction [7]. Wu et al. instead integrated SSHI with active diodes to control the inversion interval automatically; their measured circuit harvested 2.1 times the power of a full bridge with 85% conversion efficiency [8]. These circuits are useful references for a deeper next-stage timing study, rather than evidence that either active schematic tested here was already optimal. TODO: these circuits are from papers.

Chapter 4

Component Selection

The capacitor value used was 2.2 nF which . The passive detector was also swept with real transistor and diode models at underwater source levels which is discussed in the next chapter. At 230 dB, the baseline BAT54 and TIP31C/TIP32C implementation produced about 35 μ W, while the best tested combination using BSS4130A, BSS5130A, and MBR0520L produced about 56 μ W. At 250 dB, the corresponding values were about 20.5 mW and 21.4 mW. These results show that device selection matters.

Chapter 5

Underwater Acoustic Equivalent Source and Frequency Effects

5.1 Equivalent Circuit Used in the Simulations

Lutz Lampe developed frequency-domain acoustic–mechanical–electrical equivalent circuits for the underwater transducer used in this work [3, 4]. The model converts incident acoustic pressure into an electrical Thevenin source with a series equivalent branch and piezoelectric capacitance. The simulation only requires the resulting terminal-equivalent source; the internal derivation is not itself an optimization variable.

Two forms of the equivalent circuit were used during the work. The older form and the newer clamped-capacitance form have very different internal V_{eq} , R_{eq} , L_{eq} , C_{eq} , and C_p values, but when transformed consistently they present essentially the same source at the piezoelectric terminals. Consequently, the choice between the two forms does not change the circuit-design conclusions.

For example, at 42 Hz and 230 dB, the old representation gives approximately $R_{eq} = 520 \text{ M}\Omega$, $L_{eq} = 68 \text{ H}$, $C_{eq} = 579 \text{ nF}$, and $C_p = 336.5 \text{ nF}$, whereas the new representation gives approximately $R_{eq} = 0.62 \text{ }\Omega$, $L_{eq} = 29.5 \text{ nH}$, $C_{eq} = 211 \text{ nF}$, and $C_p = 125 \text{ nF}$. Despite this, both produce approximately 1.73 V open circuit at the piezoelectric port.

5.2 Numerical Check: Old and New Forms Give the Same Result

The terminal-equivalence claim was checked directly by running the same rectifier and switching simulations with both models. Representative results are in Table 5.1.

Table 5.1: Representative old/new underwater-model power comparison.

Case	Old model	New model	Difference
SEH, 230 dB	18.68 μW	18.66 μW	negligible
SEH, 250 dB	3.59 mW	3.58 mW	negligible
BJT P-SSHI, 230 dB	29.55 μW	29.50 μW	negligible
BJT P-SSHI, 250 dB	21.16 mW	21.13 mW	negligible

At 250 dB, both forms also gave $|V_{oc}| \approx 17.28 \text{ V}$ and the same terminal output impedance in the comparison calculation. The remainder of the report therefore uses whichever representation is convenient for a particular simulation and focuses on the observable terminal parameters.

5.3 Source Level

For the representative $A = 2 \times 10^{-3} \text{ m}^2$ PZT element at 42 Hz, the terminal open-circuit amplitude scales with acoustic pressure. Three useful reference points are:

Table 5.2: Representative underwater open-circuit voltage levels at 42 Hz.

Acoustic level	$ V_{oc} $
180 dB	5.46 mV
230 dB	1.73 V
250 dB	17.28 V

The 180 dB point is the regime in which semiconductor thresholds and leakage can dominate. The 230 dB point is useful for comparing practical switching and transformer matching because the open-circuit voltage is already high enough that a conventional bridge can conduct, while the source impedance remains strongly reactive.

5.4 Frequency Sweep and SSHI On-Time

A frequency sweep at 230 dB was performed while keeping $L_i = 47$ mH fixed. The maximum load powers were:

Table 5.3: Underwater frequency sweep at 230 dB with fixed SSHI inductor.

Frequency	SEH	P-SSHI	S-SSHI
1 Hz	0.0004 mW	0.004 mW	0.002 mW
10 Hz	0.0044 mW	0.054 mW	0.024 mW
100 Hz	0.043 mW	0.519 mW	0.232 mW
1000 Hz	0.439 mW	0.717 mW	0.303 mW

Test / Interface	Frequency	Optimal Rload (Ω)	Max Harvested/Load Power (mW)
SEH	1 Hz	4.64E+05	0.0004
SEH	10 Hz	1.00E+05	0.0044
SEH	100 Hz	4.64E+03	0.0434
SEH	1,000 Hz	1.00E+03	0.4393

For underwater test case
SPL_dB = 230;

Roughly power increases
proportionally to the frequency.

Test / Interface	Frequency	Optimal Rload (Ω)	Max Harvested/Load Power (mW)
PSSHI	1 Hz	1.00E+07	0.004
PSSHI	10 Hz	4.64E+05	0.054
PSSHI	100 Hz	1.00E+05	0.519
PSSHI	1,000 Hz	4.64E+03	0.717

In SSHI cases with a higher
frequency, the switch remains
closed for much longer in a
cycle (80% at 1000Hz vs 0.08%
at 1Hz). So the voltage cannot
flip cleanly resulting in less
power.

Test / Interface	Frequency	Optimal Rload (Ω)	Max Harvested/Load Power (mW)
SSSHI	1 Hz	4.64E+05	0.002
SSSHI	10 Hz	1.00E+03	0.024
SSSHI	100 Hz	1.00E+03	0.232
SSSHI	1,000 Hz	1.00E+03	0.303

Figure 5.1: Frequency sweep and optimum-load results from the underwater Simscape model.

The raw power tends to increase with frequency, but the fixed inversion inductor becomes progressively less appropriate. With $L_i = 47$ mH, Eq. (2.39) gives approximately 395 μ s. That is only about 0.08% of a half-cycle at 1 Hz but about 80% of a half-cycle at 1000 Hz. In the SSHI simulations, the switch therefore remains closed for a much larger fraction of the mechanical cycle as frequency increases. The voltage can no longer complete a clean, short inversion, and the P-SSHI waveform becomes increasingly SEH-like.

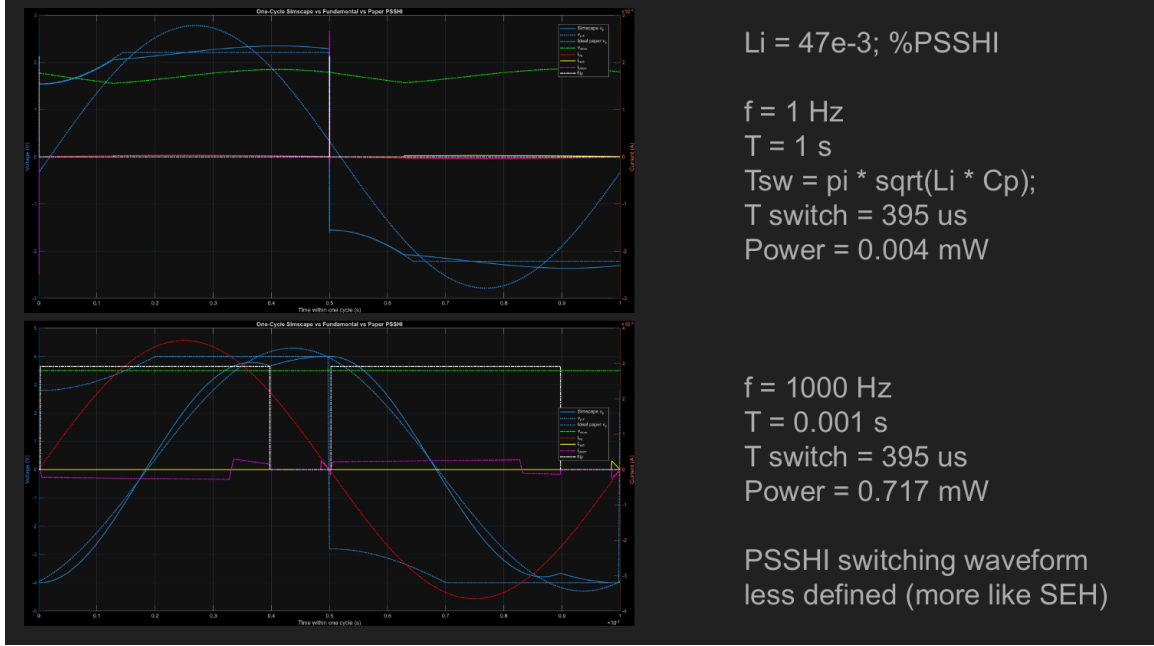


Figure 5.2: P-SSHI switching with fixed $L_i = 47$ mH at 1 Hz and 1000 Hz. The same absolute switching time occupies a much larger fraction of the high-frequency cycle.

Reducing the inversion inductance at 1000 Hz restored a short switching event. With $L_i \approx 188$ μ H, the simulated inversion time was about 25 μ s and the harvested power increased to about 5.9 mW.



Figure 5.3: High-frequency P-SSHI result after reducing L_i so that the LC inversion occupies a small fraction of the 1000 Hz cycle.

A separate same-frequency sweep showed that L_i itself was not highly sensitive while the switching interval remained short enough: P-SSHI results for 1 mH, 10 mH, 100 mH, and 1 H were approximately 0.23, 0.22, 0.25, and 0.23 mW in that test. The important quantity is therefore the ratio of T_{sw} to the mechanical period, together with switching-loop loss, not simply a single “best” inductance independent of frequency.

Chapter 6

LTspice Transformer Matching and Real-Component Sweeps

6.1 Why the Simulation Moved to LTspice

The Simscape model was useful for reproducing the analytical paper and establishing an ideal-switch upper bound. After that validation, continuing to sweep ideal switches was no longer the main need. The circuit questions had become device-level questions: which diode leakage model is acceptable, how much transformer winding resistance can be tolerated, what primary inductance is required, how sensitive is the match to frequency, and what happens when actual catalog component models are inserted.

LTspice was therefore used because it allowed much faster parameter sweeps and direct use of manufacturer-style SPICE models for diodes, BJTs, MOSFETs, inductors, and coupled windings. Ideal switching remained only as a reference where useful; the optimization itself moved toward real components.

6.2 Reactive Matching Condition for the Transformer Primary

The underwater piezoelectric terminal is strongly capacitive. Consider the piezoelectric capacitance C_p in parallel with a transformer primary magnetizing inductance L_p . Their

susceptances are

$$B_C = \omega C_p, \quad (6.1)$$

$$B_L = -\frac{1}{\omega L_p}. \quad (6.2)$$

At a chosen operating frequency, the first-order reactive match occurs when these cancel:

$$\omega C_p - \frac{1}{\omega L_p} = 0, \quad (6.3)$$

so

$$\boxed{L_p = \frac{1}{\omega^2 C_p}}. \quad (6.4)$$

This equation is a *frequency-specific shunt-resonance condition*. It is not by itself the general maximum-power-transfer theorem. It removes the dominant capacitive susceptance at one frequency so that a larger fraction of the source current can be transferred into the resistive load branch rather than circulating reactively through C_p .

For the old Lampe representation at 42 Hz, $C_p \approx 336.5$ nF, giving

$$L_p \approx 42.7 \text{ H}. \quad (6.5)$$

This explains why the low-frequency transformer problem is difficult: tens of henries of primary inductance are required before turns ratio is even considered.

6.3 Turns Ratio and Reflected Load

In this report the physical transformer turns ratio is defined as

$$n = \frac{N_s}{N_p}. \quad (6.6)$$

For an ideal transformer,

$$\frac{V_s}{V_p} = n, \quad \frac{I_s}{I_p} = \frac{1}{n}, \quad L_s \approx n^2 L_p. \quad (6.7)$$

A secondary load Z_L is reflected to the primary as

$$Z_{L,\text{ref}} = \left(\frac{N_p}{N_s} \right)^2 Z_L = \frac{Z_L}{n^2}. \quad (6.8)$$

Thus, L_p primarily controls the reactive match at the source frequency, while n and R_{load} determine the effective resistive loading seen at the primary. Some early MATLAB sweeps used a variable called `WindingRatio` with the reciprocal convention; the results in this report use the physical definition above to avoid ambiguity.

6.4 Relation to Previous Impedance-Matching Work

Transformer-based impedance matching is not new. Jabbar et al. proposed piezoelectric energy-harvester impedance matching using a step-down *piezoelectric* transformer and an inductor-less bidirectional half bridge [5]. Their motivation was closely related to the present result: a piezoelectric source has a high, capacitive impedance, and effective power transfer requires both resistive transformation and compensation of the reactive part. Their piezoelectric transformer also reduced the high harvester voltage and avoided the large magnetic component used in conventional approaches.

The magnetic-transformer version examined here is less common than SSHI for low-frequency harvesters for practical reasons. Equation (6.4) can require very large magnetizing inductance at tens of hertz. A transformer with that inductance is physically large, expensive, and usually has non-negligible winding resistance and core loss. In addition, the cancellation is narrowband: a small change in f , C_p , or L_p destroys the reactive match. P-SSHI uses a much smaller inductor only during a brief inversion interval and therefore does not require the magnetic component to remain reactively matched during the entire vibration cycle. The transformer can nevertheless be much better at a carefully tuned operating point, which is exactly what the LTspice results showed.

6.5 Direct Transformer versus P-SSHI Comparison at 230 dB

A direct LTspice comparison used the same underwater source at 42 Hz and 230 dB, corresponding to $|V_{oc}| \approx 1.73$ V. The transformer used $L_p = L_s \approx 42.68$ H with $300\ \Omega$ series resistance in each winding, while the P-SSHI branch used a 15 mH inductor with about $3.39\ \Omega$ series resistance.

Transformer vs PSSHI at f=42 and dB=230

```

req_result: 519681456.972207=5.19681e+008
ceq_result: 5.79198750091799e-007=5.79199e-007
leq_result: 67.9743345827381=67.9743
cp_result: 3.364591368864e-007=3.36459e-007 = 336.5 nF
veq_result: 79730.5444501322=79730.5
voc_result: 1.72792951790842=1.72793
f_result: 42=42
db_result: 230=230
Transformer L primary: 42.6785 H, R primary = 300 ohm
Transformer L secondary: 42.6785 H, R secondary = 300 ohm
PSSHI inductor: 15mH, Rseries = 3.39 ohm
R2 = Power across PSSHI Rload, R4 = Power across transformer Rload
Power still much higher (around 20x higher)

```

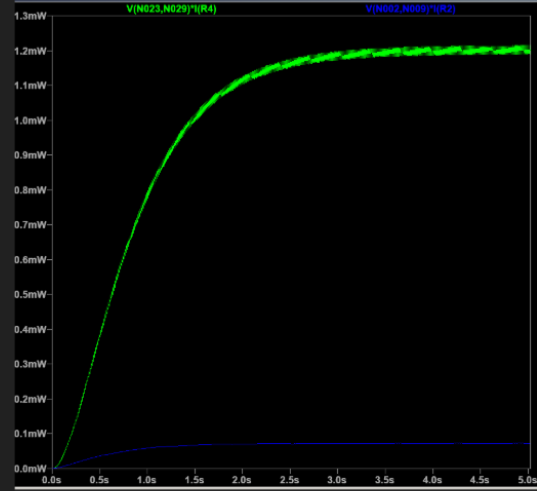


Figure 6.1: Direct LTspice comparison of transformer matching and P-SSHI at 42 Hz and 230 dB. The transformer load trace settles near 1.2 mW, while the compared P-SSHI trace is near 0.07 mW, an advantage on the order of 20 times at this tuned point.

This result is important because the transformer advantage did not appear only in the 5.46 mV case where diode forward voltage is a dominant issue. It also appeared at an open-circuit voltage of about 1.73 V. The advantage therefore came from impedance/reactive matching as well as voltage transformation.

6.6 Sensitivity to L_p and C_p

The same comparison also showed why the transformer result should not be interpreted as a universal replacement for SSHI. Small parameter changes caused large power changes.

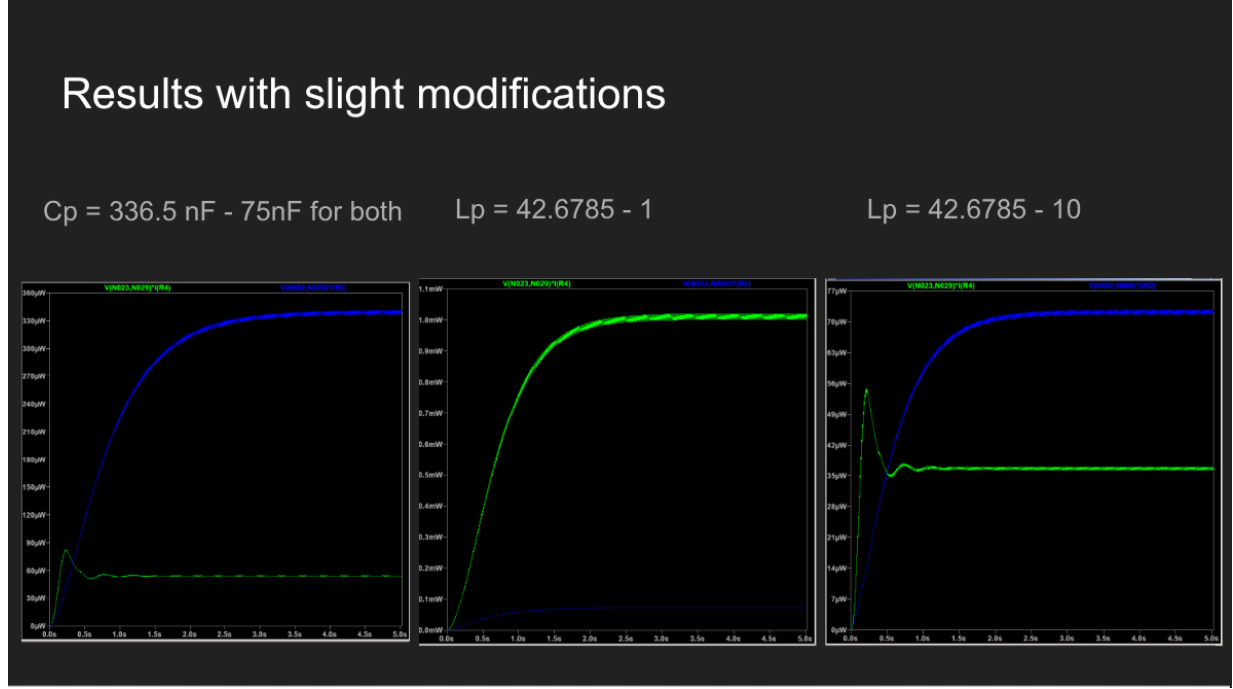


Figure 6.2: Sensitivity of the 230 dB transformer/P-SSHI comparison. Reducing the modeled C_p by 75 nF or shifting L_p away from its 42.68 H target can reduce or reverse the transformer’s advantage.

Approximate steady-state values read from these traces are:

Table 6.1: Approximate sensitivity results from the 230 dB comparison.

Modification	Transformer	P-SSHI
Nominal match	1.2 mW	0.07 mW
C_p reduced by 75 nF	55 μW	330 μW
L_p reduced by 1 H	1.0 mW	0.08 mW
L_p reduced by 10 H	36 μW	72 μW

The non-monotonic behavior is expected near a high-Q resonance: a one-henry change is small relative to 42.7 H but can already move the source away from exact cancellation, while a ten-henry shift is enough to remove most of the benefit.

6.7 Frequency and Winding-Resistance Sensitivity

Other LTspice sweeps quantified the same effect at lower source levels. With a nominal C_p -matched primary at 42 Hz, one diode/transformer configuration produced about 32.5 nW.

Changing the source to 42.1 Hz reduced the result to about 10 nW, and 43 Hz reduced it to about 0.15 nW. Adding 200 Ω primary and 100 Ω secondary winding resistance reduced the 42 Hz result to about 1.6 nW.

At 100 Hz, a 7.5 H test primary gave about 200 nW; a 1 Hz frequency error reduced that result to about 8 nW. Adding 20 Ω /10 Ω winding resistance reduced it to about 53 nW. These sweeps explain why a mathematically matched ideal transformer can look exceptional while an actual catalog transformer performs much worse.

6.8 Real Transformer Candidates and Reactive Capacitor Tuning

A high-inductance transformer close to the 42 Hz target was modeled with approximately 44 H on both windings and about 178 Ω winding resistance. It produced roughly 1.15 nW in the tested 42 Hz case, compared with roughly 160 nW for an exact idealized match. At 200 Hz, Hammond 560J winding configurations produced tens of nanowatts with their measured winding resistances, while idealized matching could reach substantially higher values.

Because exact inductance is difficult to obtain with a fixed catalog part, a parallel capacitor bank was also tested as a tuning variable. If a fixed transformer inductance is lower than the ideal matched inductance, the required total capacitance is

$$C_{\text{total}} = \frac{1}{\omega^2 L_p}, \quad C_b = C_{\text{total}} - C_p. \quad (6.9)$$

In one 200 Hz example, adding approximately 177 nF increased the simulated result from roughly 1.2 pW without the added capacitor to approximately 36.6 nW. A ± 10 nF error caused about a threefold reduction, while single-nanofarad adjustments produced much smaller changes. This result motivated the switched coarse/fine capacitor bank in the proposed PCB design.

6.9 Diode Selection in the Low-Voltage Case

At the lowest underwater source voltage, diode forward voltage and leakage both became significant. A low-forward-drop Schottky diode was not automatically best because leakage could consume current after voltage transformation. For example, in one transformed high-impedance case, a low-leakage silicon diode outperformed BAT54 despite the latter's lower forward drop. In another broader sweep, MBR0520L gave the best result at a particular

turns ratio. These observations are treated as source-specific component-selection results, not a general conclusion that one diode technology is always superior.

Chapter 7

Multiple Piezoelectric Elements and Phase Mismatch

7.1 Why Series/Parallel Connection Was Tested

Combining multiple piezoelectric elements is an attractive way to raise voltage or current. If sources are in phase, series connection can increase voltage and parallel connection can increase current. In practice, different mechanical locations or resonance frequencies can introduce phase mismatch, so direct AC combination can cause cancellation.

Du et al. experimentally studied this problem for two piezoelectric cantilevers under noisy excitation [6]. They measured about $8.4\mu\text{W}$ when each harvester used its own full-bridge rectifier and both rectifier outputs charged a common storage capacitor. Direct parallel connection before one bridge produced about $3.3\mu\text{W}$, and direct series connection produced about $2.7\mu\text{W}$. Their conclusion was that phase mismatch can make raw series/parallel connection substantially worse than separately rectifying each harvester.

7.2 Simulated Series and Phase-Shift Tests

The LTspice model was duplicated to test the same mechanism. With two identical in-phase sources connected in series ahead of the SEH bridge, the source voltages added and the output increased. Reversing one source caused strong cancellation, and imposing a 90-degree phase difference reduced the benefit relative to the in-phase case.

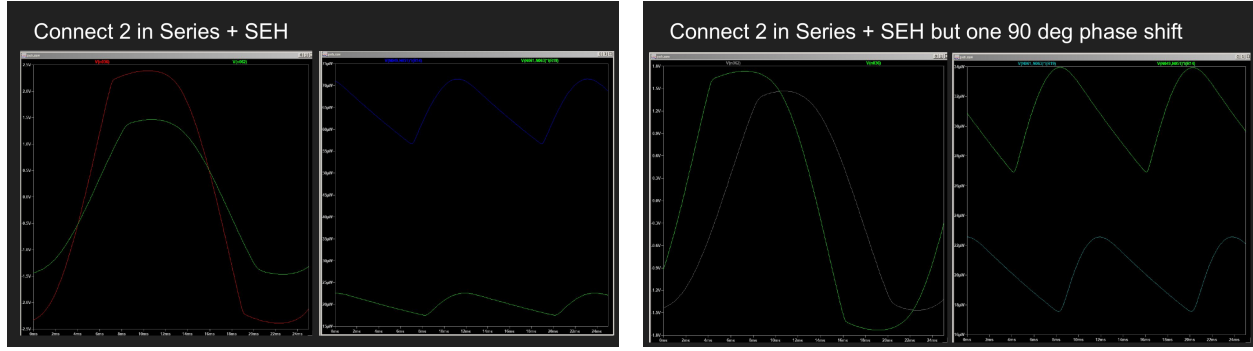


Figure 7.1: Two-source series simulations: in-phase connection (left) and a 90-degree phase mismatch (right).

The alternative was to rectify each source separately and combine only on the DC side. This prevents one AC source from directly discharging or cancelling another at a different phase.

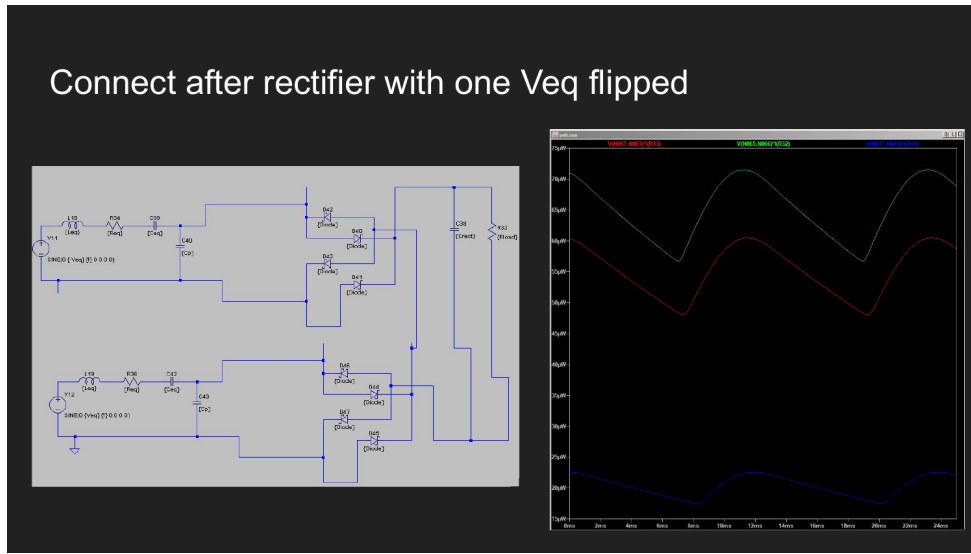


Figure 7.2: Simulation in which two piezoelectric sources are rectified separately and combined at the storage/load node, following the architecture favored by Du et al. under phase mismatch [6].

These results support a modular multi-piezo PCB in which each element can be tested either before or after rectification. The architecture should not assume phase alignment until it has been measured in the final mechanical/acoustic installation.

Chapter 8

Discussion, Hardware Direction, and Future Work

8.1 One Optimization Problem Across Regimes

The simulations form one continuous progression. The vibration benchmark established whether the nonlinear interface and impedance equations were being modeled correctly. Self-powered switching exposed timing loss. The underwater equivalent source then changed the available voltage and impedance, and LTspice exposed component losses that were hidden by ideal Simscape switches. Transformer and multi-source simulations were not separate research questions; they were attempts to solve the same interface problem after the dominant limitation changed.

At conventional benchmark voltage, P-SSHI was clearly the best of the three tested ideal interfaces. At the lowest underwater voltage, semiconductor loss and impedance transformation dominated. At 230 dB, where $V_{oc} \approx 1.73$ V, carefully matched transformer coupling could still exceed P-SSHI by a large factor, showing that the advantage was not simply a diode-threshold artifact.

8.2 Why Transformer Matching Is Powerful but Fragile

The transformer result can be understood by separating two operations. Equation (6.4) cancels the source capacitance at the target frequency; Eq. (6.8) then transforms the load resistance. When both are favorable and winding loss is small, the source sees a much more useful electrical load and power can rise sharply. But the same resonance makes the result highly sensitive to component tolerances and excitation frequency. This sensitivity is the

main reason the transformer should be treated as a calibrated narrowband interface rather than a universally superior topology.

P-SSHI has the opposite trade-off. It does not require a continuously resonant multi-henry primary and can use a much smaller inductor, but it requires accurate event timing and an efficient switching path. The self-powered BJT experiment showed that a nearly correct LC duration is insufficient if the trigger arrives almost 1 ms late.

8.3 Proposed Modular PCB

The most useful hardware implementation is therefore a modular board rather than one fixed interface. The current design direction is:

1. a piezoelectric input connector and high-impedance test points;
2. selectable SEH and P-SSHI branches so that the baseline and switched interface can be compared on the same source;
3. multiple selectable L_i values for P-SSHI timing tests;
4. a transformer branch that can be bypassed;
5. one or more transformer winding configurations or external transformer connectors;
6. a switched capacitor bank for coarse/fine reactive tuning;
7. interchangeable rectifier diode footprints or daughter boards;
8. an adjustable resistive load for measured load sweeps; and
9. connectors that allow one or multiple piezoelectric elements to be combined before or after rectification.

The first measurements should characterize V_{oc} , source impedance, frequency, and phase before selecting a fixed transformer or capacitor setting.

8.4 Limitations

The reported results are primarily simulation results. The Liang–Liao benchmark is grounded in a published experiment, but the underwater circuit values, transformer sweeps, and component selections still require bench validation with the actual transducer and magnetic

parts. Core loss, frequency-dependent inductance, parasitic capacitance, PCB leakage, cable capacitance, and acoustic/mechanical boundary conditions can all move the optimum. The 230 dB transformer/P-SSHI comparison is therefore evidence of a potentially strong matched operating point, not proof that the same ratio will be obtained from an arbitrary physical transformer.

The active zero-cross circuits also need a more systematic timing analysis. Their plotted results show that the tested versions did not beat the passive detector, but this does not show that active switching is intrinsically worse. Published active/self-powered circuits use different sensing and timing strategies [7, 8].

8.5 Future Work

The next steps are intentionally limited to experiments and circuit ideas that were not yet validated in the main results:

1. measure the actual underwater transducer’s terminal impedance, open-circuit voltage, dominant frequency content, and phase between multiple elements;
2. test the transformer and capacitor bank with measured rather than nominal inductance and DCR;
3. reproduce the 230 dB transformer/P-SSHI comparison on hardware and sweep intentional detuning around the matched point;
4. investigate better zero-cross/peak-timing circuits in more detail, including strategies from Zouari et al. and Wu et al.;
5. test active rectification, MOSFET bridge approaches, and SECE as later interface alternatives only after their quiescent power and leakage are included;
6. design and test a cold-start/energy-reservoir path if active circuitry is required at the lowest source level; and
7. compare direct series/parallel piezoelectric arrays with individual rectifiers under measured phase mismatch.

Chapter 9

Conclusions

The work answered the research question by building the interface model in layers and checking each layer against a more physical constraint. In the vibration benchmark, the simulation reproduced both impedance and power, not merely waveform appearance. The final maxima were about 0.5 mW for SEH, 1.76 mW for S-SSHI, and 2.18 mW for P-SSHI. Eq. 42 predicted the representative P-SSHI harvested power within about 0.2%, and Eq. 43 predicted the fundamental extracted power within about 3%.

Replacing ideal timing with a self-powered BJT detector reduced the P-SSHI maximum to about 1.82 mW. The measured switch duration was close to the expected half-LC interval, but the approximately 950 μ s trigger delay showed why switching phase is a separate design variable from LC duration.

For the underwater transducer, the old and new Lampe equivalent circuits were numerically interchangeable at the terminal level, so the important quantities were the resulting V_{oc} , source impedance, and frequency rather than the internal representation. At 180 dB the source was only about 5.46 mV; at 230 dB it was about 1.73 V.

The LTspice stage then showed that transformer matching can be extremely powerful when the primary inductance satisfies the reactive match and winding loss is low. At 230 dB, one tuned transformer simulation produced roughly 1.2 mW compared with roughly 0.07 mW for the compared P-SSHI configuration. However, detuning L_p , changing C_p , or adding winding resistance could reduce the output by orders of magnitude. This is why magnetic-transformer matching is less common than SSHI in low-frequency piezoelectric harvesters despite its high simulated peak performance: it is narrowband and demands a large, low-loss primary inductance.

The most defensible conclusion is therefore conditional. P-SSHI is the strongest validated nonlinear interface in the conventional vibration benchmark. Transformer/reactive

matching can be stronger for a stable underwater source when the source capacitance, frequency, magnetizing inductance, turns ratio, and winding resistance are all controlled. A practical interface should therefore be tunable and modular, and its final selection should follow measurement of the actual transducer rather than assumption of a single universally optimal topology.

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